ETHIOPIAN SEED ASSOCIATION · WEBSITE

Deployment Guide

How to run the ESA website on your own computer, build it for production, and publish it on Vercel with your own domain.

Project: **esa-website** · Next.js 16 · React 19 · Tailwind CSS 4 · Recharts
Version 1.0 · October 2026

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Project at a glance

Item	Location
Project folder	`Documents\1SPACE\Website\esa-website`
Site text, members, news, publications	`src/lib/content.ts`
National seed data for the charts	`src/lib/seed-data.ts`
Pages (one folder per URL)	`src/app/`
Photos and logos	`public/images/`
Contact form endpoint	`src/app/api/contact/route.ts`

Verified: on this machine, `npm install` and `npm run build` complete successfully and generate 101 pages.

1. What you need

Tool	Version	Why
Node.js	20.9 or newer (22 or 24 LTS recommended)	Runs Next.js. Includes `npm`. Download from nodejs.org.
A terminal	PowerShell, Command Prompt or Git Bash	To type the commands in this guide.
Git (for section 4)	Any recent version	To push the code to GitHub. Download from git-scm.com.
GitHub account (section 4)	Free	Stores the code; Vercel deploys from it.
Vercel account	Free Hobby plan to test; Pro for an organisation	Hosts the site. Sign up at vercel.com.

Check what you have installed:

```
node -v      # should print v20.9.0 or higher
npm -v
git --version
```

> **Plan note:** Vercel's Hobby plan is for personal, non-commercial use. For ESA's official site, use a Pro team owned by ESA so the account stays in ESA's name (requirement C9).

2. Run it locally (development)

The development server reloads the page automatically whenever you save a file. Use it while editing content.

1. **Open a terminal in the project folder**

```
cd "C:\Users\Administrator\Documents\1SPACE\Website\esa-website"
```

2. **Install dependencies (first time only)**

This downloads the libraries into a `node_modules` folder. It takes 1–3 minutes.

```
npm install
```

3. **Start the development server**

```
npm run dev
```

You should see `Ready` and `Local: http://localhost:3000`.

4. **Open the site**

Go to **http://localhost:3000** in your browser. Try the member directory, publications and the seed data dashboard.

5. **Stop the server**

Press `Ctrl + C` in the terminal.

> **Port already in use?** Run on another port: `npm run dev -- --port 3100`, then open http://localhost:3100.

> **View on your phone:** the terminal also shows a `Network:` address such as http://192.168.x.x:3000. Open it on a phone connected to the same Wi-Fi. Windows Firewall may ask you to allow Node.js.

3. Production build on your computer

This is exactly what Vercel runs. Doing it locally first catches errors before you deploy.

1. **Build**

   ```
   npm run build
   ```

   A successful build ends with a route list and a line like `Generating static pages (101/101)`. TypeScript errors stop the build and show the file and line.

2. **Start the production server**

   ```
   npm start
   ```

   Open **http://localhost:3000**. This is the fast, optimised version of the site.

3. **Optional: check quality**
   In Chrome, open DevTools (F12) → **Lighthouse** → Mobile → Analyze. The Phase 1 target (N2) is a mobile score of 85 or more.

Hosting on a local server or office PC

To keep the site running on a Windows machine on ESA's network, build once, then run `npm start`. To keep it running after the terminal closes, use a process manager:

```
npm install -g pm2
pm2 start npm --name esa-website -- start
pm2 save
```

> For a public website, Vercel (next sections) is simpler and safer than hosting on an office PC: it gives HTTPS, backups of every deployment, and a global CDN without maintenance.

4. Deploy on Vercel: GitHub method (recommended)

With this method, every change you push to GitHub is deployed automatically, and every version can be rolled back.

1. **Create an empty repository on GitHub**
   On github.com click **New repository**, name it `esa-website`, choose **Private**, and do not add a README. Click **Create repository**.

2. **Push the project from your computer**

```
cd "C:\Users\Administrator\Documents\1SPACE\Website\esa-website"
git init
git add .
git commit -m "ESA website v1"
git branch -M main
git remote add origin https://github.com/YOUR-ACCOUNT/esa-website.git
git push -u origin main
```

   The included `.gitignore` already excludes `node_modules` and `.next`, so only source files are uploaded.

3. **Import the project into Vercel**
   Sign in at vercel.com → **Add New…** → **Project** → **Continue with GitHub** → authorise access → choose `esa-website` → **Import**.

4. **Check the settings**
   Vercel detects Next.js automatically. Leave the defaults:

Setting	Value
Framework Preset	Next.js
Root Directory	`./` (if you pushed the parent *Website* folder instead, set it to `esa-website`)
Node.js Version	22.x or 24.x (Project Settings → Build and Deployment)

5. **Add the environment variable**
   Under **Environment Variables** add `SITE_URL` = your final address, for example `https://esa-website.vercel.app`. (See section 6.)

6. **Click Deploy**
   The build takes about 1–2 minutes. When it finishes you get a live address like **https://esa-website.vercel.app**.

5. Deploy on Vercel: CLI method

Use this to deploy straight from your computer without GitHub. Good for a quick preview link to share with ESA reviewers.

1. **Install the Vercel CLI**

```
npm install -g vercel
```

2. **Log in**

```
vercel login
```

Choose your login method; a browser window opens to confirm.

3. **Deploy a preview**

```
cd "C:\Users\Administrator\Documents\1SPACE\Website\esa-website"
vercel
```

Answer the questions: *Set up and deploy?* **Y** · choose your scope/team · *Link to existing project?* **N** · project name **esa-website** · directory **./** · keep the detected Next.js settings. You get a **preview URL**.

4. **Add the environment variable**

```
vercel env add SITE_URL production
```

Paste the production address when asked.

5. **Deploy to production**

```
vercel --prod
```

Preview deployments

Every `vercel` run (or every GitHub branch / pull request) gets its own URL. Use these for ESA review rounds without touching the live site.

Rollback

In the Vercel dashboard → **Deployments**, open any earlier deployment → ⋯ → **Promote to Production** (or **Instant Rollback**).

6. Custom domain and environment variables

Environment variables

Name	Example	Used for
`SITE_URL`	`https://www.ethiopianseedassociation.org`	Page metadata, `sitemap.xml` and `robots.txt`. No trailing slash.

After changing an environment variable in Vercel, redeploy (Deployments → latest → **Redeploy**) for it to take effect.

> The domain above is an example only. Use the domain ESA registers in its own name.

Connect ESA's domain

1. **Add the domain in Vercel**
   Project → **Settings** → **Domains** → enter the domain (e.g. `www.your-domain.org`) → **Add**.

2. **Create the DNS records at your domain registrar**
   Vercel shows the exact records to add. Typically:

Type	Name	Value
A	`@` (root domain)	the IP address Vercel shows
CNAME	`www`	the CNAME target Vercel shows

   Always copy the values from your Vercel dashboard, as they can change.

3. **Wait for verification**
   DNS changes take from a few minutes to 48 hours. Vercel issues the HTTPS certificate automatically once the domain points to it.

4. **Update `SITE_URL`**
   to the new domain and redeploy.

7. Updating the site

Typical content changes

To change…	Edit
Board members, secretariat	`board` and `secretariat` in `src/lib/content.ts`
Add or remove a member	`members` list in `src/lib/content.ts`
Add a news item or event	`news` list in `src/lib/content.ts`; put the photo in `public/images/news/`
Add a publication	`publications` list; include title, type, year, language, size in bytes and URL
New season of seed data	Add a row to `NATIONAL_Q` in `src/lib/seed-data.ts` (values in quintals)
Contact details	`org.contact` in `src/lib/content.ts`

Publish the change

```
# 1. check it locally
npm run dev
# 2. make sure it builds
npm run build
# 3a. GitHub method: push, and Vercel deploys automatically
git add .
git commit -m "Add March 2027 seed forum news"
git push
# 3b. CLI method
vercel --prod
```

> Requirement F10 asks that staff can update content without a developer. A headless CMS (for example Sanity, Strapi or Payload) can later replace `content.ts` so that editors work in a web form instead of code.

8. Before launch checklist

Based on the Phase 1 Requirement Analysis Report. Tick each item before pointing the public domain at the site.

✓	Ref.	Item
☐	C1	Every ESA fact on the site verified against an official document and logged in the Source-of-Truth Register.
☐	C3	Leader and staff photos and 80–150 word biographies supplied with written consent.
☐	C4	Members' consent to publish name, logo, region, crops and contact details.
☐	F9	Contact form connected to ESA's shared mailbox. `src/app/api/contact/route.ts` currently validates and accepts messages but does not send email; connect an email service (e.g. Resend, SendGrid or SMTP) and store its API key as a Vercel environment variable.
☐	C7	Privacy notice, accessibility statement and terms approved by the legal adviser.
☐	C9	Domain and Vercel team registered in ESA's name with two ESA administrators.
☐	N4	Two-step sign-in turned on for every Vercel and GitHub admin account.
☐	N2	Lighthouse mobile score of 85 or more on the home page.
☐	N7	`SITE_URL` set to the real domain; `/sitemap.xml` submitted in Google Search Console.
☐	—	Remove the "Prototype for Phase 2 review" line from the footer (`src/components/site-footer.tsx`).

9. Troubleshooting

Problem	Fix
`'npm' is not recognized`	Node.js is not installed or the terminal was open before installing. Install Node.js, then open a new terminal.
`Port 3000 is in use`	Another app uses the port. Run `npm run dev -- --port 3100`, or close the other app.
Build fails with a TypeScript error	The message names the file and line. Fix it, then run `npm run build` again. A missing comma or quote in `content.ts` is the usual cause.
Fonts fail to download during build	The build fetches Google Fonts. Check your internet connection or proxy and retry.
Image not showing	Paths are case-sensitive on Vercel. `/images/news/Photo.JPG` and `/images/news/photo.jpg` are different files.
Strange errors after updates	Delete the `.next` folder and run `npm run build` again. If needed, also delete `node_modules` and run `npm install`.
Vercel build fails but local build works	Open the failed deployment → **Build Logs**. Check the Node.js version (22.x or 24.x) and that all files, including images, were committed.
Domain shows "Invalid Configuration"	The DNS records don't match yet. Re-check them against Vercel's Domains page and allow time to propagate.

Command summary

```
npm install          # install dependencies (first time)
npm run dev          # development server, http://localhost:3000
npm run build        # production build
npm start            # run the production build locally
vercel               # preview deployment
vercel --prod        # production deployment
```